

Hybrid resolution refinement and the rodeo algorithm for eigenstate preparation: a demonstration on the pairing Hamiltonian

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The efficient preparation of correlated eigenstates is a central bottleneck for quantum simulation: variational quantum eigensolvers (VQE) suffer from exponentially vanishing gradients, adiabatic state preparation requires evolution times that diverge when the spectral gap closes, and projective filters such as quantum phase estimation and the rodeo algorithm converge exponentially only when the input state already has a sizeable overlap with the target eigenstate. We demonstrate, end to end and on a single controlled testbed — the constant-pairing Hamiltonian — a hybrid pipeline that resolves these tensions: a cheap low-resolution VQE state is prolonged into a larger model space, adiabatically *refined* across resolutions, and finally polished by the rodeo algorithm. Resolution refinement keeps the path gapped, so the adiabatic time scales only as $T \sim 1/\Delta E$ and the overlap p with the high-resolution ground state rises toward unity; the rodeo algorithm then removes the residual error exponentially, with a post-selected infidelity $1 - \mathcal{F}_M \lesssim \frac{1-p}{p} 2^{-M}$ in the number of cycles M and a post-selection acceptance that stays of order p . For four particles refined up to four levels ($k = 4$, eight qubits) the refined input ($p \simeq 0.998$) is filtered to a fidelity $1 - \mathcal{F} \sim 10^{-6}$ within one or two cycles, while an unrefined prolonged guess ($p \simeq 0.87$) still reaches $\sim 10^{-4}$ in a handful of cycles. All gate-level building blocks (UCCSD state preparation, the prolongation embedding, and the ancilla-based rodeo cycle) are implemented explicitly and validated against exact diagonalisation. The construction is the pairing-model realisation of the EIGEN-Q proposal and a concrete template for hardware demonstrations on mid-circuit-measurement platforms such as the Magne neutral-atom quantum computer.

I. INTRODUCTION

Determining the low-lying eigenvalues and eigenvectors of a many-body Hamiltonian is one of the grand challenges of quantum many-body theory, where the Hilbert-space dimension grows exponentially with the number of constituents. Quantum computers offer new routes to eigenvalue estimation and eigenstate preparation, but the state-of-the-art methods each face an obstacle at scale. The variational quantum eigensolver (VQE) [1] is limited by barren plateaus — gradients that vanish exponentially with system size [2]. Adiabatic state preparation requires evolution times that diverge if the gap between the tracked state and the rest of the spectrum closes along the path [3]. Projective spectral filters — quantum phase estimation [4] and the rodeo algorithm (RA) [5, 6] — converge exponentially in the number of cycles, but only once the initial state has a significant squared overlap p with the target eigenstate; their cost grows as $1/p$, and p itself typically falls exponentially with system size.

These limitations are complementary, which suggests a hybrid strategy. *Resolution refinement* [7] bootstraps

an eigenstate across model-space scales: an eigenstate of a coarse (low-resolution) Hamiltonian is lifted by a prolongation operator into a finer space and adiabatically evolved to the fine-resolution eigenstate. Because refinement preserves the low-energy spectral structure, the path stays gapped and the required adiabatic time scales only as the inverse physical gap (times a slowly growing system-size factor), delivering an output with $p \approx 1$. That output is exactly the high-overlap input the rodeo algorithm needs to converge exponentially. This is the core idea of the EIGEN-Q pipeline: (i) prepare a low-resolution eigenstate; (ii) prolong and adiabatically refine to a final time T^* ; (iii) apply the rodeo algorithm to reach the target fidelity and observables with exponential convergence.

In this work we realise the full pipeline end to end on a transparent and exactly solvable testbed, the constant-pairing Hamiltonian familiar from nuclear and condensed-matter many-body theory [13]. Working on a model whose exact spectrum is available at every stage lets us validate each component quantitatively: the gate-level UCCSD coarse solver against full configuration interaction (FCI) and coupled-cluster doubles (CCD); the prolongation and adiabatic refinement against the exact high-resolution eigenstate; and the rodeo filter against the analytic success-probability formulas of Ref. [6].

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All circuits are implemented explicitly as rotation and CNOT gates (and an ancilla-based rodeo cycle), so the demonstration doubles as a reference implementation for hardware.

II. THE PAIRING MODEL AND ITS QUBIT ENCODING

We study the constant-pairing Hamiltonian

$$H = \delta \sum_{p=0}^{k-1} p \sum_{\sigma=\uparrow\downarrow} n_{p\sigma} - \frac{g}{2} \sum_{p,q=0}^{k-1} a_{p\uparrow}^\dagger a_{p\downarrow}^\dagger a_{q\downarrow} a_{q\uparrow}, \quad (1)$$

with k doubly degenerate single-particle levels of spacing δ (set to $\delta = 1$) and a pairing strength g . The interaction scatters time-reversed pairs between levels; it conserves the seniority (the number of unpaired particles), so the ground state of N paired particles lives entirely in the seniority-zero subspace. We map level p and spin σ to qubit $2p + \sigma$ (so a k -level problem uses $2k$ qubits) and encode the fermionic operators through the Jordan–Wigner transformation. The Hartree–Fock (HF) reference fills the lowest $N/2$ levels, with $E_{\text{HF}}(N=4, g) = 2 - g$.

Throughout we use $N = 4$ particles as the running example. The fixed- N sector dimension is $\binom{2k}{N}$, growing from 1 at $k = 2$ (a single filled determinant) to 70 at $k = 4$ and 1820 at $k = 8$; the seniority-zero ground-state problem is correspondingly compact, which makes the model an ideal multi-resolution laboratory. “Basis refinement” here means *increasing k at fixed N* : the number of single-particle levels (and qubits) grows while the particle number is held fixed.

III. THE HYBRID PIPELINE

A. Coarse-state preparation: gate-level UCCSD–VQE

The coarse eigenstate is prepared with a unitary coupled-cluster ansatz restricted to singles and doubles (UCCSD), Trotterised into a product of excitation circuits acting on the HF reference,

$$|\Psi(\theta)\rangle = \prod_{\mu} e^{\theta_{\mu}(\hat{\tau}_{\mu} - \hat{\tau}_{\mu}^{\dagger})} |\text{HF}\rangle. \quad (2)$$

Through the Jordan–Wigner map each anti-Hermitian generator becomes a sum of commuting Pauli strings,

so every factor compiles into a basis change, a CNOT ladder, a single $R_z(\theta)$ rotation, and the uncompute; a single excitation yields two Pauli-rotation sub-circuits and a double excitation yields eight. We optimise with the exact analytic gradient of the product ansatz (the naive two-term parameter shift is incorrect beyond the first factor, because later factors acquire frequency-one cross terms). Because the pairing force does not couple to seniority-changing single excitations, all singles amplitudes vanish to machine precision and the solver is effectively a *doubles* theory, mirroring the classical CCD structure.

B. The prolongation operator

The prolongation P embeds a low-resolution state into the high-resolution space. For basis refinement it maps every many-body basis state of the k_{low} space to the *same* occupation pattern in the k_{high} space, leaving the new single-particle levels empty. In canonical Jordan–Wigner ordering this is gate-free; for lattice refinement the analogous operator distributes each occupied coarse site over the neighbouring fine sites with two single-qubit rotations and two CNOTs per site per spatial direction, i.e. $O(N_{\text{sites}})$ gates per dimension [7]. The prolonged state $|\Phi_0\rangle = P|\psi_{\text{low}}\rangle$ satisfies $\langle\Phi_0|H_{\text{high}}|\Phi_0\rangle = E_{\text{low}}$ exactly, and already provides a high-quality first guess for the high-resolution ground state.

C. Resolution refinement (adiabatic phase)

The prolonged state is refined by adiabatically following the interpolated, energy-shifted Hamiltonian

$$H(s) = \cos^2\left(\frac{\pi s}{2}\right) P(H_{\text{low}} - \mu)P^\dagger + \sin^2\left(\frac{\pi s}{2}\right) (H_{\text{high}} - \mu), \quad s = \frac{t}{T}, \quad (3)$$

evolved by a Suzuki–Trotter product over total time $T \leq T^*$. The shift μ (here $\mu = E_{\text{low}} + 0.6$) places the

tracked low-energy states below the large null space of $P^\dagger$, so the level followed from $s = 0$ to $s = 1$ is the

high-resolution ground state. The instantaneous spectrum stays smooth and gapped along the entire path; consequently the adiabatic time scales as $T \sim 1/\Delta E$ rather than the $O(1/\Delta E_{\min}^2)$ of generic adiabatic preparation, and the overlap $p = |\langle \Psi_{\text{high}} | \Phi(T) \rangle|^2$ rises toward unity.

D. The rodeo algorithm

The rodeo algorithm [5, 6] removes the residual error left by the adiabatic tail. One cycle acts on the system register plus a single ancilla (reused across cycles by mid-circuit measurement): a Hadamard on the ancilla, the controlled evolution e^{-iHt_k} of the system, a phase gate $P(E t_k)$ on the ancilla (which effectively shifts $H \rightarrow H - E$), a second Hadamard, and an ancilla measurement. Post-selecting the ancilla in $|0\rangle$ applies the non-unitary filter $\frac{1}{2}(\mathbb{I} + e^{iEt_k}e^{-iHt_k})$ to the system: an eigenstate $|E_j\rangle$ acquires the factor $e^{i(E-E_j)t_k/2} \cos[(E_j - E)t_k/2]$. For an eigenstate input the probability that all M ancilla measurements return 0 is therefore [6]

$$P_{0M}(E | \{t_k\}) = \prod_{k=1}^M \cos^2 \left[\frac{t_k}{2} (E_{\text{obj}} - E) \right], \quad (4)$$

and averaging the t_k over a Gaussian of RMS width σ gives

$$P_{0M}(E) = \left[\frac{1 + e^{-(E_{\text{obj}} - E)^2 \sigma^2 / 2}}{2} \right]^M. \quad (5)$$

On resonance ($E = E_{\text{obj}}$) the success probability is unity at every cycle, while off-resonant weight is suppressed geometrically toward the background 2^{-M} (the arithmetic mean of $\cos^2$ is $1/2$). Scanning E thus locates eigenvalues as peaks whose width $\sim 1/\sigma$ sets the energy resolution; fixing E at the target eigenvalue turns the algorithm into a state-preparation filter.

The controlled evolution is the only system-size-dependent ingredient. On hardware e^{-iHt} is realised by a Suzuki–Trotter decomposition over the $2k$ qubits [10–12]; for the single-qubit benchmark of Ref. [6] it reduces to a Z – Y – Z Euler rotation $R_{\hat{n}}(\theta) = e^{-i\theta \hat{n} \cdot \vec{\sigma}/2}$ implemented through the Qiskit native gate. In the simulations below we apply the controlled evolution exactly within the conserved N -particle sector, isolating the algorithmic behaviour from Trotter error (quantified in Sec. V).

a. Convergence criterion. With E tuned to the target eigenvalue E_0 , the all-zero acceptance probability of an input $|\psi\rangle = \sum_k c_k |E_k\rangle$ after M cycles is

$$P_M = p + \sum_{k \neq 0} |c_k|^2 \left[\frac{1}{2} (1 + e^{-(E_k - E_0)^2 \sigma^2 / 2}) \right]^M, \quad p = |c_0|^2. \quad (6)$$

The target weight p is preserved at every round while off-resonant components decay geometrically, so $P_M \rightarrow p$

TABLE I. Ground-state energies at $g = 1$, $N = 4$, for the constant-pairing Hamiltonian. FCI is the exact N -sector result; CCD is coupled-cluster doubles; UCCSD is the gate-level circuit of Sec. III A. “#CNOT” is the two-qubit gate count of the UCCSD state preparation.

k	qubits	FCI	CCD	UCCSD	#CNOT
2	4	1.000000	1.000000	1.000000	0
3	6	0.794697	0.794697	0.794697	272
4	8	0.635548	0.630443	0.636987	1312
6	12	0.372664	0.350058	0.378719	–
8	16	0.145559	0.097263	0.157856	–

and the post-selected fidelity obeys

$$\mathcal{F}_M = \frac{p}{P_M} \simeq \left[1 + \frac{1-p}{p} 2^{-M} \right]^{-1}, \quad 1 - \mathcal{F}_M \lesssim \frac{1-p}{p} 2^{-M}. \quad (7)$$

Reaching fidelity $1 - \delta$ therefore needs only $M \gtrsim \log_2[(1-p)/(p\delta)]$ cycles, and since the expected number of repetitions to obtain an accepted run is $\sim 1/P_M \rightarrow 1/p$, the total controlled-evolution effort scales as $O(|\log \delta|/(p\epsilon))$ at energy resolution ϵ [5]. This is efficient precisely when resolution refinement supplies $p \approx 1$.

IV. RESULTS

A. Coarse solver: UCCSD versus exact references

Table I lists the ground-state energies at $g = 1$ for $N = 4$ as the basis grows from $k = 2$ to $k = 8$. The gate-level UCCSD circuit reproduces the operator-level (statevector) UCCSD to machine precision and stays variational ($E_{\text{UCCSD}} \geq E_{\text{FCI}}$), tracking FCI to $\sim 10^{-3}$ – 10^{-2} , whereas CCD over-binds. Figure 1 compares all solvers (exact, CCD, statevector and gate-level UCCSD, and a hardware-efficient ansatz) as a function of k . The hardware-efficient ansatz is exact at $k = 2$ and reaches FCI at $k = 3$ only after a multi-start optimisation, and struggles at $k = 4$ — the symmetry-blind landscape that motivates the physically structured UCCSD circuit. Figure 2 shows that the same UCCSD solver remains a tight variational bound all the way to $k = 8$ (sixteen qubits), at the cost of a CNOT count that grows steeply (1312 CNOTs already at $k = 4$).

B. Resolution refinement: overlap and energy

At $N = 4$ the low-resolution ($k = 2$) space is a single filled determinant, so the prolonged state is the high-resolution HF reference, with first-guess overlaps 0.917 for refinement to $k = 3$ and 0.868 to $k = 4$. Adiabatic evolution along the gapped path of Eq. (3) drives both quantities toward their exact targets: the overlap rises to

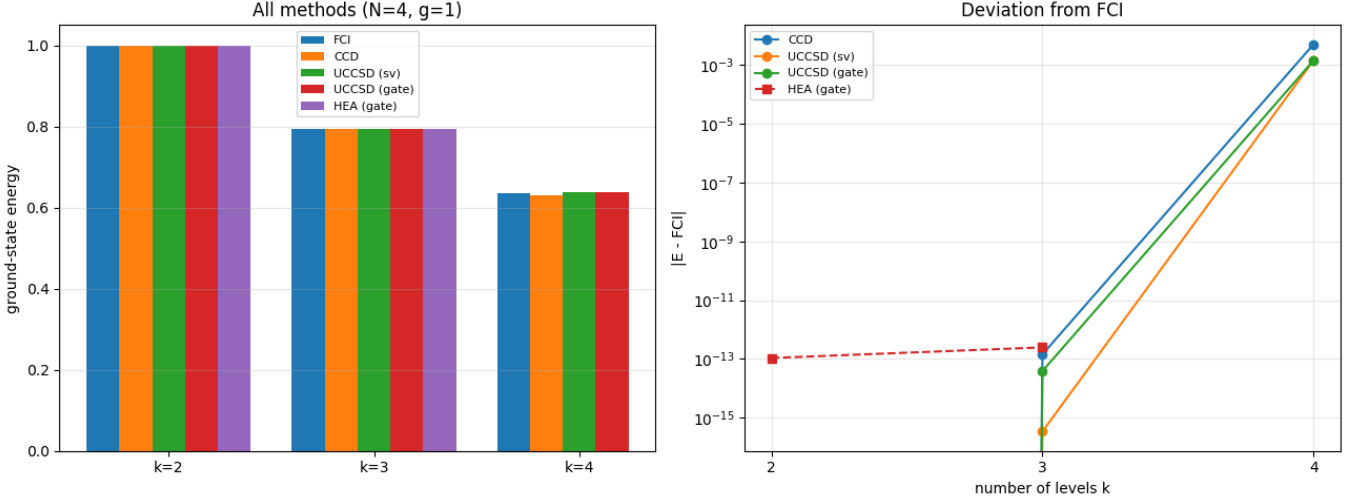


FIG. 1. Ground-state energy at $N = 4$, $g = 1$ obtained with all solvers (left) and their deviation from FCI (right), versus the number of levels k . The structured UCCSD circuits track FCI from above; CCD over-binds; the hardware-efficient ansatz matches FCI for $k \leq 3$ but loses accuracy at $k = 4$.

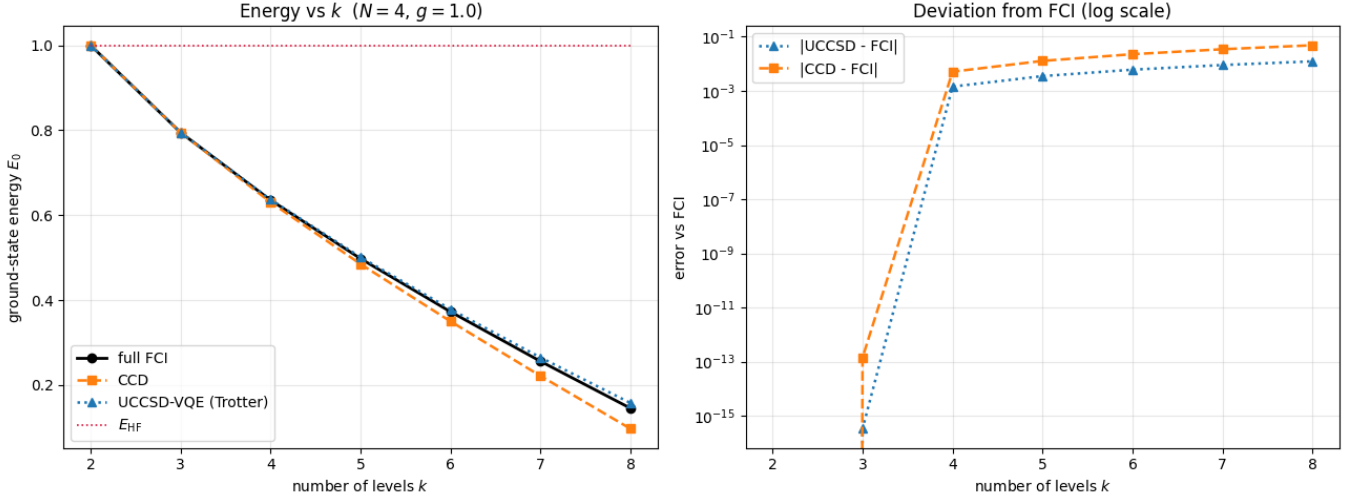


FIG. 2. Scalability of the Trotterised UCCSD-VQE: ground-state energy (left) and deviation from FCI (right) from $k = 2$ to $k = 8$ ($N = 4$, $g = 1$). The same solver remains variational and tight as the basis — and hence the qubit count — grows.

$\simeq 0.999$ and the energy falls from $E_{HF} = 1$ to the FCI value from above — 0.7964 for $k = 3$ (FCI 0.794697) and 0.6397 for $k = 4$ (FCI 0.635548) at $T = 30$ (Fig. 4). The instantaneous spectrum stays open along the whole path (Fig. 3, right), confirming the favourable $T \sim 1/\Delta E$ timescale; the larger $k = 4$ space starts lower in overlap and converges more slowly, exactly as expected from the system-size dependence of the refinement.

C. Rodeo filtering of the refined state

We verified the rodeo primitive in two ways: the explicit ancilla circuit agrees with the compact filter $\frac{1}{2}(\mathcal{K} + e^{iEt}e^{-iHt})$ to 10^{-16} , and the all-zero probabili-

ties reproduce Eqs. (4)–(5), with the Gaussian-averaged single-cycle factor matching $\frac{1}{2}(1 + e^{-(\Delta E \sigma)^2/2})$ to three digits.

The state fed to the filter is the resolution-refined $k = 4$ state of Sec. IV B, with overlap $p = 0.998$ with the exact ground state; for contrast we also use the unrefined prolonged guess Φ_0 ($p = 0.868$). Scanning the target energy E (Fig. 5) produces a peak at the ground-state energy; a coarse scan ($\sigma = 2$) resolves it from the excited band, and a fine scan ($\sigma = 8$) locates it at $E = 0.6352$ against the exact $E_0 = 0.635548$. Starting from the refined state, essentially all spectral weight sits in the single ground-state peak; starting from Φ_0 , the small residual weight on excited states is visible as secondary structure above

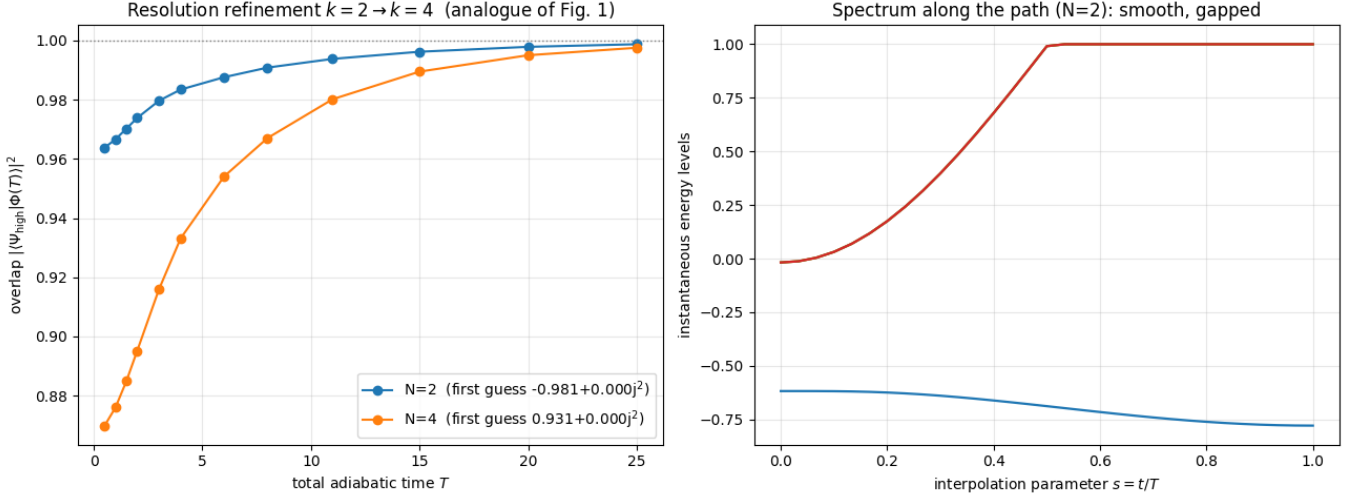


FIG. 3. Resolution refinement $k = 2 \rightarrow k = 4$. Left: overlap of the evolved state with the exact high-resolution ground state versus the total adiabatic time T , for $N = 2$ and $N = 4$. Right: the instantaneous spectrum along the interpolation parameter $s = t/T$ stays smooth and gapped, the condition for efficient refinement.

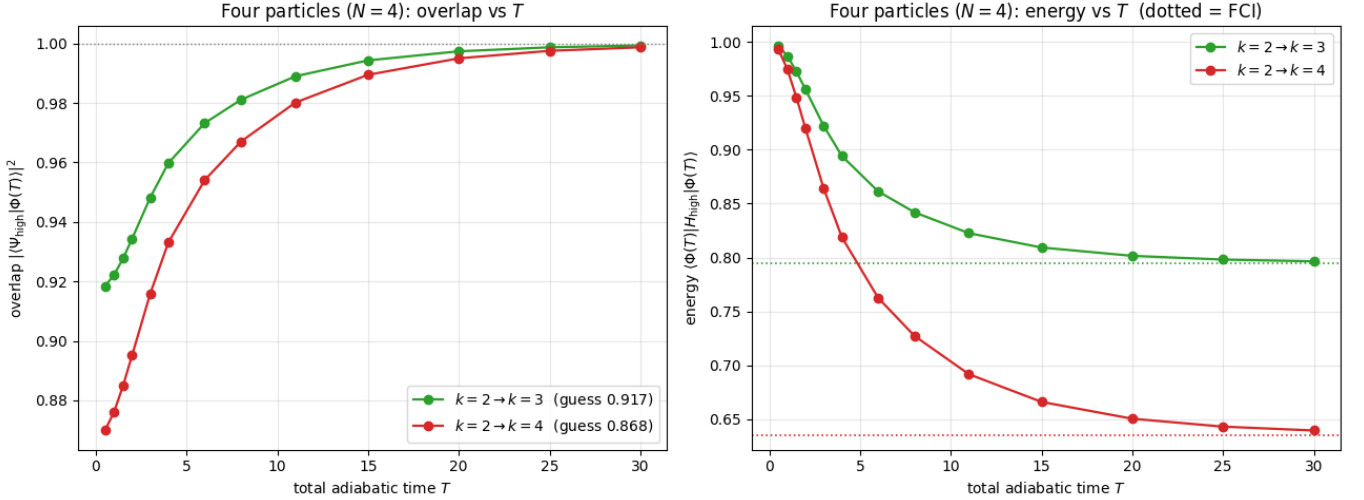


FIG. 4. Four-particle refinement up to $k = 4$: overlap (left) and physical energy $\langle \Phi(T) | H_{\text{high}} | \Phi(T) \rangle$ (right) versus T for $k_{\text{high}} = 3, 4$. The energy approaches the FCI value (dotted) from above while the overlap approaches unity, both on a $\sim 1/\Delta E$ timescale.

the 2^{-M} background.

Fixing $E = E_0$ and applying M cycles (Fig. 6), the post-selected fidelity rises exponentially and the cumulative acceptance settles at $\approx p$, in quantitative agreement with Eqs. (6)–(7). The refined input reaches an infidelity $1 - \mathcal{F} \simeq 5 \times 10^{-6}$ within one to two cycles; even the cruder prolonged input reaches $\simeq 1.3 \times 10^{-4}$ in a few cycles, while its acceptance holds at 0.868. The dotted reference lines $\frac{1-p}{p} 2^{-M}$ bracket the measured infidelities, confirming the convergence criterion. This is the concrete pay-off of the pipeline: resolution refinement supplies the high overlap that turns the rodeo algorithm from a $1/p$ -expensive filter into a few-cycle polish.

V. DISCUSSION AND OUTLOOK

The pairing model lets us exhibit the entire EIGEN-Q pipeline with every stage benchmarked against exact results: a gate-level UCCSD coarse solver, a gate-free prolongation, a gapped adiabatic refinement with $T \sim 1/\Delta E$, and a rodeo filter with $1 - \mathcal{F}_M \lesssim \frac{1-p}{p} 2^{-M}$. The two algorithms are genuinely complementary — refinement provides the large overlap $p \approx 1$ that the rodeo algorithm requires, and the rodeo algorithm removes the slow adiabatic tail that refinement alone leaves behind.

Moving to hardware changes only the controlled evolution. On a device e^{-iHt} is Trotterised over the $2k$ qubits,

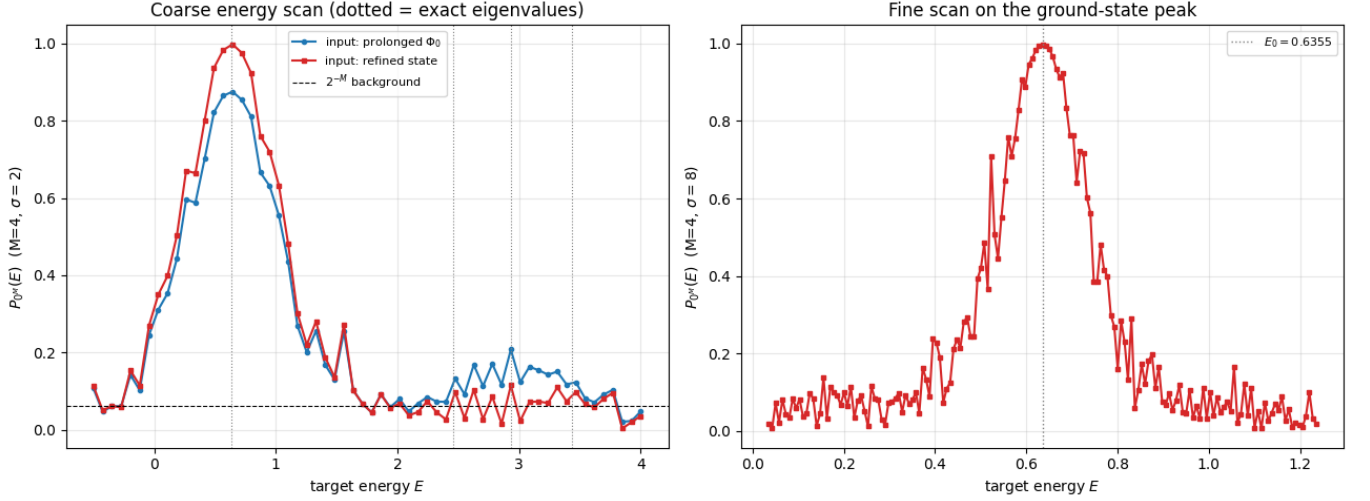


FIG. 5. Rodeo energy scan ($M = 4$). Left: coarse scan ($\sigma = 2$); the dotted vertical lines mark exact eigenvalues and the dashed line the 2^{-M} background. Right: fine scan ($\sigma = 8$) on the ground-state peak, located at $E = 0.6352$ (exact $E_0 = 0.6355$).

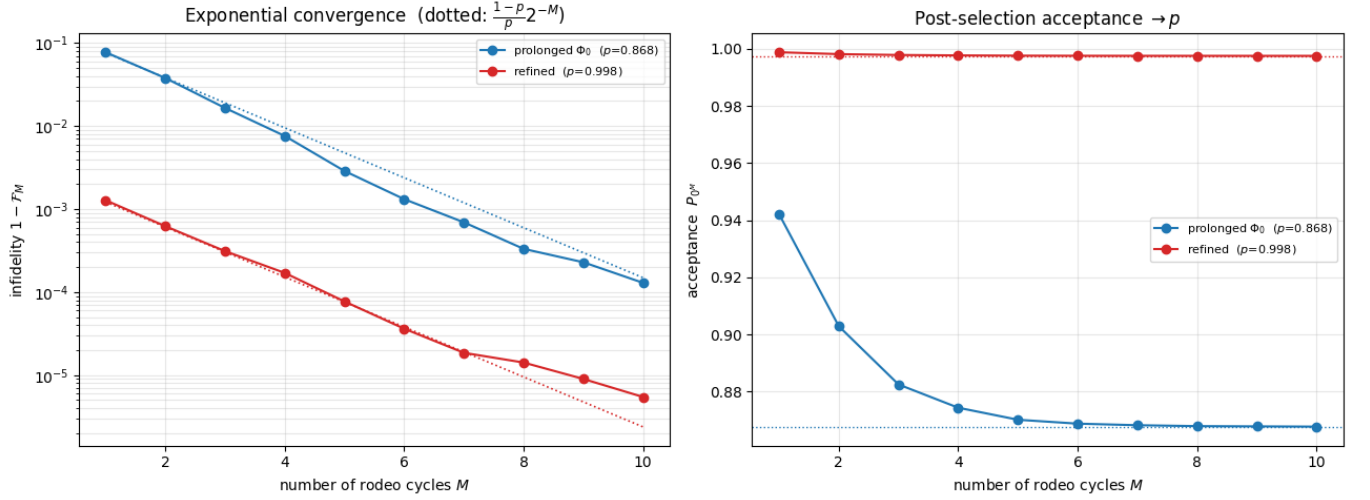


FIG. 6. Rodeo eigenstate preparation at $E = E_0$, averaged over random time sets. Left: post-selected infidelity $1 - \mathcal{F}_M$ versus the number of cycles M (dotted: the bound $\frac{1-p}{p}2^{-M}$). Right: the post-selection acceptance converges to the input overlap p .

which introduces an effective Hamiltonian whose eigenvalues differ from the exact ones at a controllable order in the step size; as noted in Ref. [6], the rodeo algorithm filters the eigenstates of this effective Hamiltonian, and its self-correcting design (small unitary errors broaden the energy window rather than biasing the estimate) makes it unusually robust to such errors. The mid-circuit measurement and deep coherent circuits required by the two phases are supported by neutral-atom architectures such as the Magne system, making the pipeline a natural target for a hardware demonstration.

Beyond the pairing model, the same construction applies to the targets of the EIGEN-Q proposal — light and medium-mass nuclei benchmarked against coupled-cluster and in-medium similarity renormalization group

calculations, molecular systems toward chemical accuracy, and Hubbard models refined from a coarse to a fine lattice — with a closed-loop control layer that can adapt the refinement schedule from the rodeo acceptance statistics read out mid-circuit. The present results establish, on a fully controlled example, that the hybrid of resolution refinement and the rodeo algorithm prepares correlated eigenstates with exponential accuracy at a cost set by an overlap that the refinement itself makes large.

CODE AVAILABILITY

The complete implementation — pairing Hamiltonian, gate-level UCCSD-VQE, prolongation, adiabatic refine-

ment, and the ancilla-based rodeo algorithm — is provided as a single Jupyter notebook that reproduces every figure in this article.

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